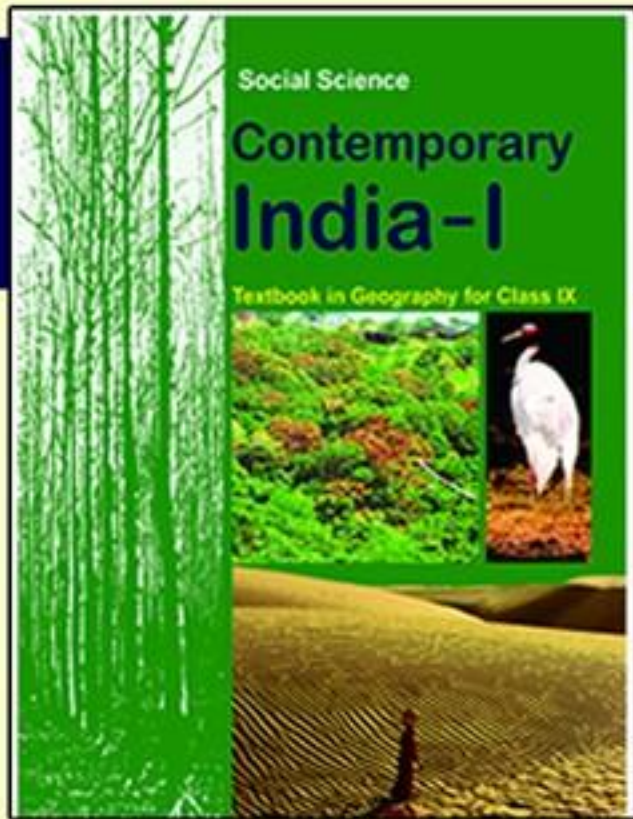




Learning **Space**



NCERT BASICS



GEOGRAPHY

MAJOR DOMAINS OF THE EARTH

- LITHOSPHERE
- ATMOSPHERE
- HYDROSPHERE
- BIOSPHERE

MODULE - 5





MAIN COMPONENTS OF EARTH

LITHOSPHERE
HYDROSPHERE
ATMOSPHERE
BIOSPHERE

The solid portion of the Earth.

Water in its all forms.

- Troposphere
- Stratosphere
- Mesosphere
- Thermosphere
- Exosphere

Narrow contact
between land,
water and air.



1. LITHOSPHERE

The solid portion comprises of the rocks of Earth's crust and thin layers of soil.



1. LITHOSPHERE

SPECIFIC ASPECTS ABOUT ASIA

DO YOU KNOW?

- Asia is the largest of the world's continents.
- It has roughly 60 percent of the total population of the world.
- It makes up the eastern portion of the Eurasian supercontinent.

These mountain ranges forms part of the conventional boundary between the regions of Europe and Asia.

URAL MOUNTAINS

ASIA

TROPIC OF CANCER PASSES THROUGH 8 STATES IN INDIA!

TROPIC OF CANCER



SPECIFIC ASPECTS ABOUT EUROPE

EUROPE

Tropic of Capricorn, Equator, Tropic of Cancer do not pass through Europe.

DO YOU KNOW?

- ✓ Europe is the second-smallest continent.
- ✓ Europe is a peninsula of the Eurasian supercontinent
- ✓ It is bordered by the Arctic Ocean to the north, the Atlantic Ocean to the west, and the Mediterranean, Black, and Caspian Seas to the south.

1. LITHOSPHERE

SPECIFIC ASPECTS ABOUT AFRICA



NILE RIVER

DO YOU KNOW?

- Africa is the second largest continent
- It covers about one-fifth of the total land surface of Earth.
- The continent is bounded on the west by the Atlantic Ocean, on the north by the Mediterranean Sea, on the east by the Red Sea and the Indian Ocean, and on the south by the mingling waters of the Atlantic and Indian oceans.

SAHARA DESERT

Tropic of Cancer (23 1/2° N)

Equator (0°)

South Atlantic Ocean

Tropic of Capricorn (23 1/2° S)

Indian Ocean

1. LITHOSPHERE

SPECIFIC ASPECTS ABOUT NORTH AMERICA AND SOUTH AMERICA



DO YOU KNOW?

- North America is the third-largest continent.
- In the far north, the continent stretches halfway around the world, from Greenland to the Aleutians.
- But at Panama's narrowest part, the continent is just 50 kilometers across.
- South America is the fourth-largest continent.
- It extends from the Gulf of Darién in the northwest to the archipelago of Tierra del Fuego in the south.

1. LITHOSPHERE

SPECIFIC ASPECTS ABOUT AUSTRALIA

DO YOU KNOW?

- ✓ Australia, the smallest continent and one of the largest countries on Earth.
- ✓ It lies between the Pacific and Indian oceans in the Southern Hemisphere.
- ✓ Australia's capital is Canberra.



1. LITHOSPHERE

SPECIFIC ASPECTS ABOUT ANTARCTICA

DO YOU KNOW?

- ✓ Antarctica is unique among the continents for being almost totally covered by glacier ice.
- ✓ The greater portion of Antarctica lies within the Eastern Hemisphere.
- ✓ It is 'the highest continent on Earth,' because of the thickness of its ice sheets.

SOUTH POLE

ANTARCTICA

2. HYDROSPHERE

ARCTIC OCEAN

IMPORTANT ASPECTS

- More than 71% of the Earth is covered with water.
- More than 97% of the Earth's water is found in oceans.

WHY DOES THE WATER IN OCEANS MOVE?

Characteristics like *Temperature, Salinity, Density, Rotation of the Earth and the External Forces like of the Sun, Moon and the Winds* influence the movement of ocean water.

➤ The 3 chief movements of oceans:

✓ Waves

✓ Tides

✓ Ocean Currents.

ARCTIC OCEAN

GREENLAND

Baffin Bay

ALASKA
USA

CANADA

UNITED STATES

NORTH PACIFIC
OCEAN

NORTH ATLANTIC
OCEAN

SOUTH PACIFIC
OCEAN

BRAZIL

SOUTH ATLANTIC
OCEAN

ANTARCTIC OCEAN

INDIAN
OCEAN

AUSTRALIA

HOW DO YOU DIFFERENTIATE BETWEEN WAVES TIDES AND OCEAN CURRENTS

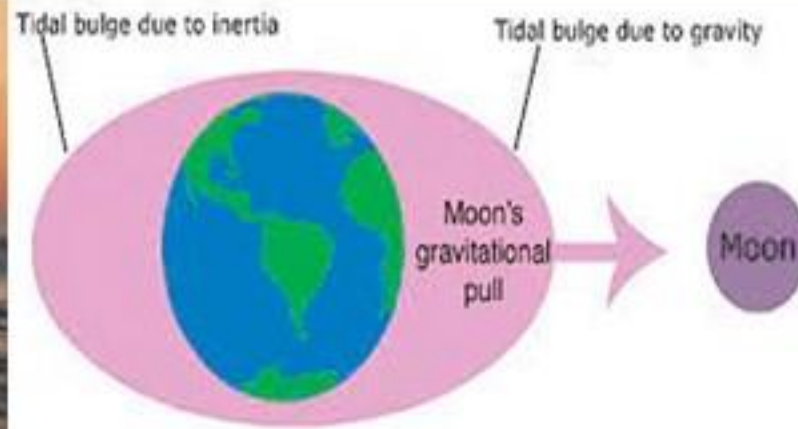
WAVES

Waves are essentially the movement of water molecules within the ocean, and are restricted to the surface layers of our oceans and seas.



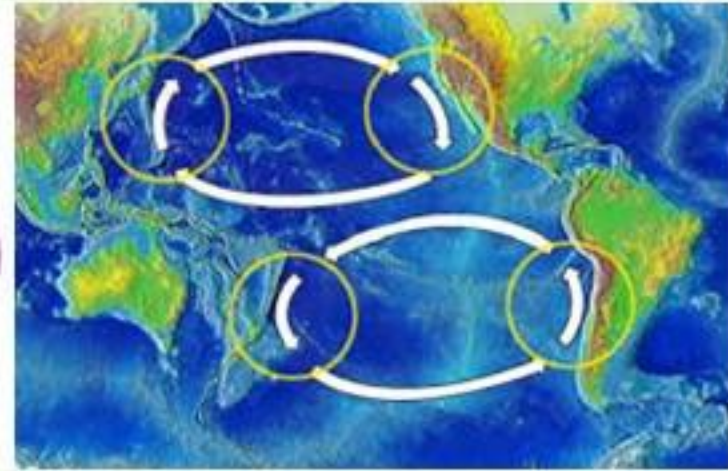
TIDES

The rhythmic rise and fall of ocean water twice a day, mainly due to the attraction of the Sun and the Moon.



OCEAN CURRENTS

Ocean currents are the continuous, predictable, *directional* movement of seawater driven by gravity, wind, and water density.



3. ATMOSPHERE

SIMPLE FACTS ABOUT THE ATMOSPHERE

- Earth's atmosphere is so much more than the air we breathe.
- Earth's atmosphere stretches from the surface of the planet up to as far as 10,000 Kms.

The majority of Earth's atmosphere is not oxygen.

- Earth's atmosphere is composed of about:
 - ✓ 78 percent Nitrogen
 - ✓ 21 percent Oxygen
 - ✓ 0.9 percent Argon
 - ✓ 0.1 percent other gases.
- *Other Gases* include:
 - ✓ Trace amounts of Carbon Dioxide, Methane, water vapor, and Neon.

3. ATMOSPHERE

5 LAYERS

i. TROPOSPHERE

- It is the lowest layer of the Earth's atmosphere, extending from the surface to about 16-20 kilometers above.
- Approximately 75 to 80 percent of the mass of the entire atmosphere is contained here.
- Temperatures in the troposphere decrease with altitude.
- The top of the troposphere, is called the **tropopause**.



3. ATMOSPHERE

5 LAYERS

ii. STRATOSPHERE

- Level of Earth's atmosphere, extending from 10 to 50 kilometers above the surface of the Earth.
- A high concentration of *ozone*, a molecule composed of three atoms of oxygen, makes up the ozone layer here.
- This ozone is responsible for the temperature increase in altitude.
- The top of the stratosphere is called the **stratopause**.



3. ATMOSPHERE

5 LAYERS



iii. MESOSPHERE

- ✓ About 50-85 kilometers above the Earth's surface.
- ✓ Thick enough so that meteors will burn up as they pass through the mesosphere, creating what we see as "shooting stars."
- ✓ Temperatures decrease in the mesosphere with altitude.
- ✓ In fact, the coldest temperatures in the atmosphere are near the top of the mesosphere.
- ✓ The upper boundary of the mesosphere is called the **mesopause**.

3. ATMOSPHERE

5 LAYERS

A common definition of space is known as the Kármán Line, an imaginary boundary 100 kilometers above mean sea level.



iv. THERMOSPHERE

- ✓ It is located above the mesopause.
- ✓ Layer of the earth's atmosphere located between 85 and 600 kilometers above the Earth's surface.
- ✓ Solar radiation makes the upper regions of the thermosphere very hot.

3. ATMOSPHERE 5 LAYERS



v. EXOSPHERE

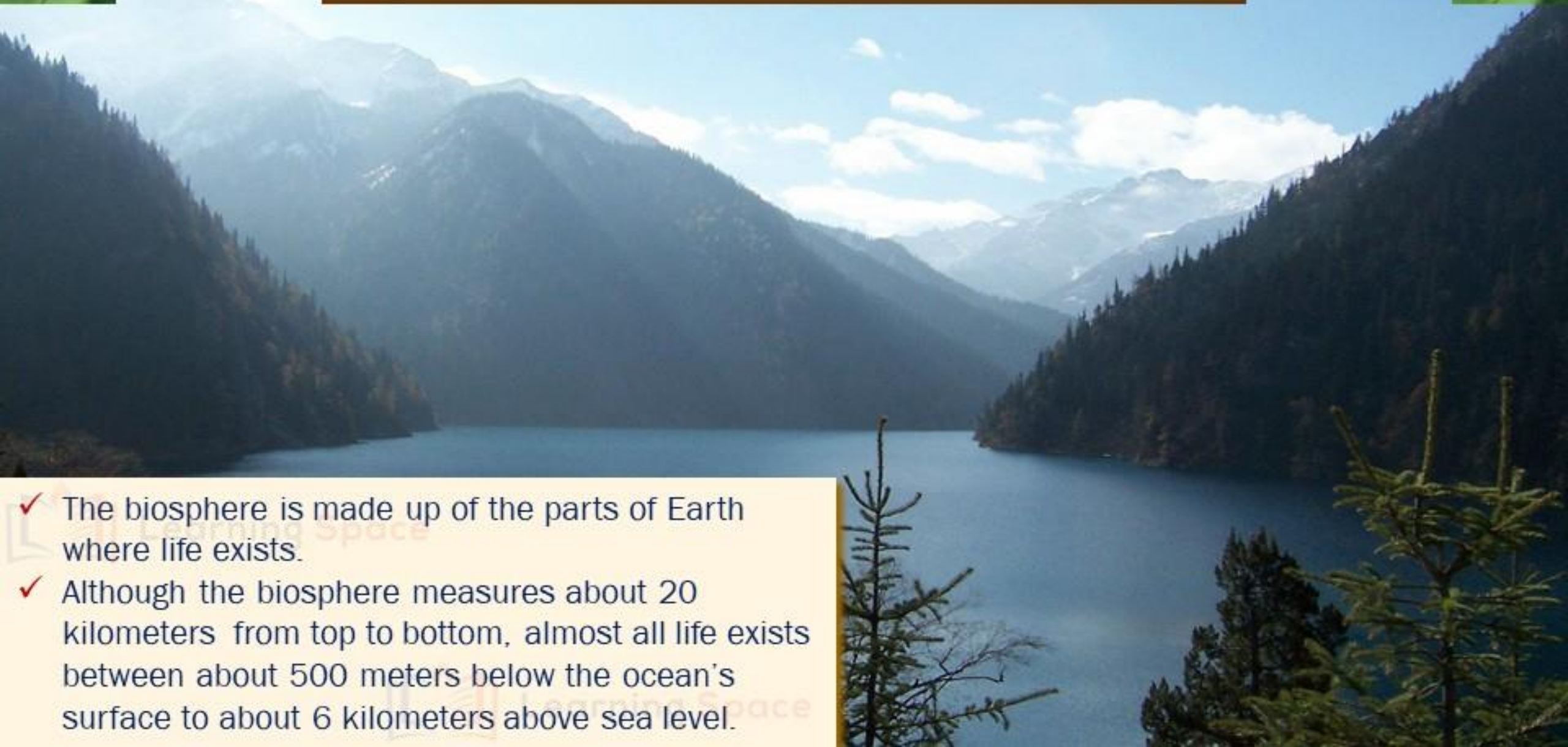
- ✓ Beginning at an altitude of about 600 kilometers above the earth's surface.
- ✓ Considered to be outer space.
- ✓ The pull of earth's gravity is so small here that molecules of gas escape into outer space.



4. BIOSPHERE



THE BIOSPHERE OVERLAPS ALL THE 3 SPHERES!



- ✓ The biosphere is made up of the parts of Earth where life exists.
- ✓ Although the biosphere measures about 20 kilometers from top to bottom, almost all life exists between about 500 meters below the ocean's surface to about 6 kilometers above sea level.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

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